

Networking Fundamentals and Hands-on Checks.

- OSI model (7 layers) → conceptual model to understand networking step-by-step
- TCP/IP model (4 layers) → practical model used in real systems.

	OSI layer	TCP/IP layer	What it does.
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L7	Application	Application	User-facing protocols (HTTP, DNS)
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L6	Presentation	Application	Data format, encryption
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L5	Session	Application	Session management
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L4	Transport	Transport	End-to-end communication
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L3	Network	Internet	Routing using IP
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L2	Data link	Link	Mac® addressing
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L1	Physical	Link	cables, signals.
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- Where things sit in stack?
- 1) IP → Internet layer
- 2) TCP/UDP → Transport layer
- 3) HTTP/HTTPS → Appⁿ layer
- 4) DNS → Appⁿ layer

• Hands-on checklist

1. Identity -
hostname -I → shows system IP address
2. Reachability -
ping google.com → checks if host is reachable & latency.
3. Path -
tracert google.com → shows path packets take to
4. Ports →
ss -tulpp → list listening ports & services.
5. Name Resolution →
nslookup google.com → resolves domain → IP
6. HTTP check -
curl -I https://google.com → Fetches HTTP headers only
7. Connection snapshot -
netstat -an | head → shows active connections.

• Reflection -

→ faster command when something is broken?
ping → quickest way to check basic connectivity.

→ If DNS fails → check which layer?
Application layer → DNS issue.

→ If HTTP 500 error?
Appⁿ layer issue

→ To follow-up checks in real incident.
systemctl status <services> → check service health.

ss -tulpn → verify port is listening.